

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

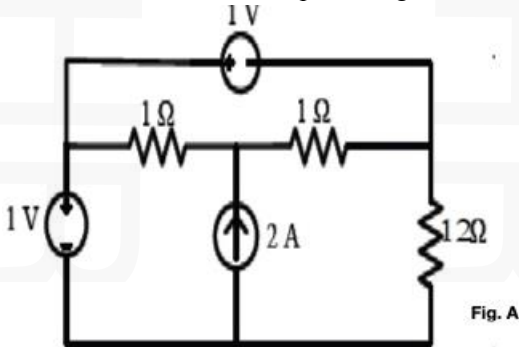
B.Tech Degree S6 (S, FE) / S6 (PT) (S, FE) Examination December 2024 (2019 Scheme)

Course Code: EET 308**Course name: COMPREHENSIVE COURSE WORK**

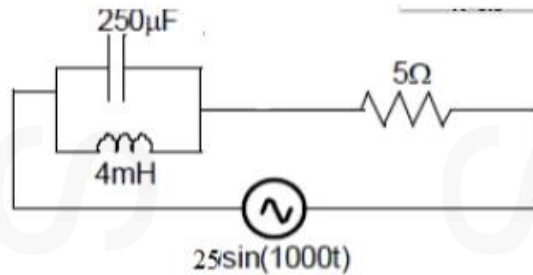
Max. Marks: 50

Duration: 1Hour

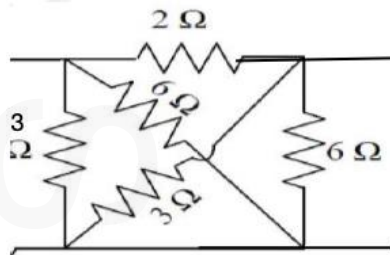
- Instruction** (1) Each question carries one mark. No negative marks for wrong answers
(2) Total number of questions: 50
(3) All questions are to be answered. Each question will be followed by 4 possible answers of which only ONE is correct.
(4) If more than one option is chosen, it will not be considered for valuation.

1. Reciprocity theorem is applicable to ?
a) linear and bilateral circuits only. b) linear and unilateral circuits only. c) All electrical circuits. d) Non-linear circuits only.
2. Calculate the current in Amps through the $12\ \Omega$ resistor?

Fig. A
a) 0.4 b) 0 c) 0.6 d) 1
3. What is an undamped electric circuit?
a) Pure Resistive b) Pure capacitive c) Inductive-Capacitive d) Resistive-Inductive-capacitive
4. In Fig. A what is Thevenin's impedance across the terminals of the $12\ \Omega$ resistor?
a) $2\ \Omega$ b) $1/2\ \Omega$ c) $0\ \Omega$ d) $3\ \Omega$

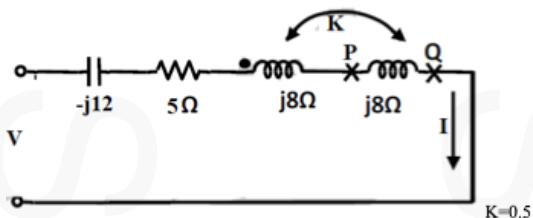
5. Find the rms current through the $5\ \Omega$ resistor ?



- a) $5 \angle 0^\circ$ b) 0 c) $5 \angle 90^\circ$ d) $5 \angle -90^\circ$
6. Using superposition, what is the power delivered in watts by the 2A source in Fig. A?
- a) 0 b) 1 c) 2 d) 3
7. Which is the best condition for symmetry and reciprocity for a network using T parameters?
- a) $A^2 - BC = 1$ b) $A = C$ c) $B = C$ d) $AD = BC$
8. What is the transfer impedance z_{12} in the given circuit?



- a) $2/3\ \Omega$ b) $2\ \Omega$ c) $3\ \Omega$ d) $6\ \Omega$
9. What is not true about Thevenin's and Norton's equivalent circuits?
- a) They are duals of each other b) Applicable only to linear circuits c) Z_{TH} is the same for both d) One can be obtained by source transformation of the other
10. The ratio of total reactance of the circuit when P is the other dotted terminal to when it is Q?



- a) -3 b) 3 c) $1/3$ d) $-1/3$

- 11 What is the function of a buffer circuit?
- a) To provide an output equal to its input b) To clean up the input c) To clean up the output d) To provide an output that is inverted from that on the input
- 12 Simplify $Y = AB' + (A' + B)C$. ?
- a) $AB' + C$ b) $AB + AC$ c) $A'B + AC'$ d) $AB + A$
- 13 Which method of error-detecting codes enables to find double errors in Digital Electronic devices?
- a) Parity method b) Check sum c) Odd-Even d) Bit generation method
- 14 Add the two BCD numbers: $1001 + 0100 = A$?
- a) $A = 10101111$ b) $A = 1010000$ c) $A = 00010011$ d) $A = 00101011$
- 15 Logic circuitry is used to detect _____
- a) Underflow b) Overflow c) MSD d) LSD
- 16 What are the the most fundamental gates in MOS logic family?.
- a) AND, NAND b) NAND, NOR c) OR, NOR d) OR, NOTR
- 17 A modulus-10 counter must have _
- a) 10 flip-flops b) 4 flip-flops c) 2 flip-flops d) Synchronous clocking
- 18 In a 4-bit Johnson counter sequence, there are a total of how many states or bit patterns?
- a) 1 b) 2 c) 4 d) 8
- 19 What is the preset condition for a ring shift c
- a) All FFs cleared to 0 b) All FFs set to 1 c) A single 0, the rest 1 d) A single 1, the rest 0
- 20 A bidirectional 4-bit shift register is storing the nibble 1110. Its input is LOW. The nibble 0111 is waiting to be entered on the serial data-input line. After two clock pulses, the shift register is storing _____
- a) 1110 b) 0111 c) 1011 d) 1001
- 21 A 240 V, dc shunt motor draws 15 A while supplying the rated load at a speed of 80 rad/s. The armature resistance is 0.5Ω and the field winding resistance is 80Ω . What is the net voltage across the armature resistance when plugging
- a) 6V b) 234V c) 240V d) 474V
- 22 In a DC machine, which of the following statements is true about Compensating Winding (CW) and Interpole Winding (IW)? ?

1200EET308122402

- a) CW neutralises armature reaction; IW produces residual flux b) CW neutralises armature reaction; IW improves commutation c) CW improves commutation; IW produces residual flux d) CW improves commutation; IW neutralises armature reaction
- 23 Which DC generator can be used as a booster in a DC power system?
a) Shunt b) Series c) Shunt d) Cumulative compound
- 24 The armature of a 220 V DC shunt motor is wound in lap to _____
a) Avoid sparking b) Reduce cost c) Reduce loss d) Increase power output
- 25 Which is the dc motor that can give zero speed regulation at full load without a controller?
a) Differential compound b) Series c) Shunt d) Cumulative compound
- 26 In a transformer, zero voltage regulation at full load is
a) possible at leading power factor load b) possible at unity power factor load c) Not possible d) possible at lagging power factor load
- 27 Two transformers are to be operated in parallel such that they share load in proportion to their kVA ratings. The rating of the first transformer is 500 kVA and its pu leakage impedance is 0.05 pu. If the rating of second transformer is 250 kVA, its pu leakage impedance is?
a) 0.2 b) 0.1 c) 0.05 d) 0.025
- 28 28. A single-phase transformer has no-load loss of 64 W, as obtained from an open circuit test. When a short-circuit test is performed on it with 90% of the rated currents flowing in its both LV and HV windings, measured loss is 81 W. At what load does the transformer have maximum efficiency?
a) 50% b) 60% c) 80% d) 88.8%
- 29 29. The equivalent circuit of a transformer has leakage reactances , X_1 , X_2 and magnetising reactance X_M . Their magnitudes satisfy:
a) $X_1 \gg X_2 \gg X_M$ b) $X_1 \ll X_2 \ll X_M$ c) $X_1 \simeq X_2 \gg X_M$ d) $X_1 \simeq X_2 \ll X_M$
- 30 30. A single-phase transformer with a turns ratio 1:2 is connected to a pure resistive load. The secondary and magnetising currents are 1A. Neglecting core losses and leakage reactances the primary current is
a) 3 A b) 2.24A c) 2A d) 1.41A
- 31 Determine the even component of the signal: $x(t) = \cos(t + 45^\circ) + \sin t$.
a) $\sin 2t$ b) $\sin t$ c) $\cos(t + 45^\circ)$ d) $\cos t$
- 32 32. For an energy signal _____
a) $E=0$ b) $P=0$ c) $E = \infty$ d) $P = \infty$
- 33 What is the period of the following signal, $x(t) = 5 \cos(24 t + 60^\circ)$?
a) 12 b) 24π c) $1/12$ d) $1/24\pi$

34 How are $x(t)$ and $y(t)$ related?



- a) $Y(t) = \int x(t) dt$ b) $Y(t) = dx(t)/dt$ c) $Y(t) = x(t/2)$ d) $Y(t) = x(t-1)$

35 Given $f(t) = [\cos(3t)]^2$. The value of $L\{f(t)\}$ is _____

- a) $s^2/(s^2+36)^2$ b) $36/(s^2+36)^2$ c) $(s/(s^2+9))^2$ d) $1/2[1/s + s/(s^2+36)]$

36 Find the initial value of $f(t)$ if $F(s) = \frac{\omega}{(s+\alpha)^2 + \omega^2}$

- a) ∞ b) 0 c) 1 d) -1

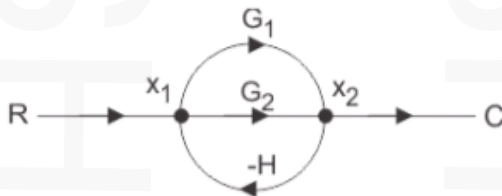
37 Bandwidth of the gate function is _____

- a) τ Hz b) $1/\tau$ Hz c) 2τ Hz d) $2/\tau$ Hz

38 Find the Z-transform of $u[-n]$

- a) 1 b) $z/(z-1)$ c) $z/(z+1)$ d) $-1/(z-1)$

39 Find the transfer function of the following system using Mason's formula:



- a) $G_1/(1+G_2H)$ b) $G_2/(1+G_1H)$ c) $G_1/(1+G_1H)$ d) $(G_1+G_2)/(1+G_1H+G_2H)$

40 Effect of feedback on sensitivity is minimum in:

- a) Open loop systems b) Closed loop systems c) Both OL and CL systems d) Neither OL nor CL systems

41 Which power plant can handle both base load and peak load operations effectively?

- a) Hydroelectric b) Thermal c) Nuclear d) Diesel

42 A power system has a maximum load of 20 MW. Annual load factor is 50%. The reserve capacity of plant in MWH is _____ if the plant capacity factor is 40%.

- a) 3.75 b) 5 c) 46.75 d) 8.75

43 Ferranti effect is very prominent in occurrence in what type of transmission line?

- a) Long, unloaded b) Short, loaded c) Long, loaded d) Short, unloaded

- 44 What will happen when the characteristic impedance of a transmission line is equal to the load impedance _____
- a) Load will reject all the given energy b) Given energy is lost as line loss c) Resonance d) Load will absorb all the given energy
- 45 An overhead transmission line has a span of 282.8 metres the conductor weighing 0.2 kg/m. What will be the maximum sag if the working tension is 2000 kg.
- a) 1m b) 0.5m c) 1.2 m d) 0.33 m
- 46 In extra High Voltage lines, which conductor material is used for ground wire provided above the line conductors for protection against lightning?
- a) ACSR b) Hard drawn copper c) Cadmium copper d) Stranded galvanised steel
- 47 The surge impedance of multiple conductor lines as compared to single line is _____
- a) Higher b) Lower c) No change d) Varies according to the length
- 48 If an industrial consumer consumes 50 kW for 4 hours, 300 kW for 15 hours and 60 kW for 5 hours daily and the tariff rate is Rs 5/kWh. What is the energy consumption in a month of 30 days in lakhs of rupees?
- a) 7.5 b) 7.75 c) 4.7 d) 4.75
- 49 What of the below mentioned statements are incorrect as compared to the HVDC system?
- a) Reactive power compensation b) Distance limitation c) More corona loss d) Back to back connection possible
- 50 To increase the power despatch of an existing transmission line install _____
- a) Series capacitors b) Shunt capacitors c) Series reactors d) Shunt reactors